



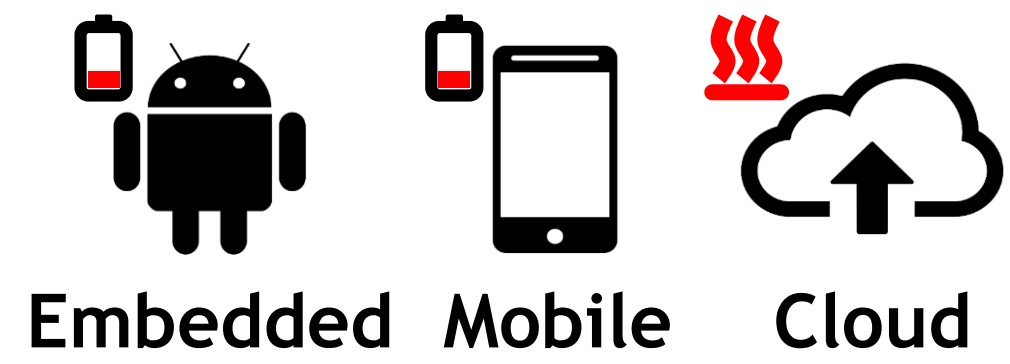
# An Architecture-Level Energy and Area Estimator for Processing-In-Memory Accelerator Designs



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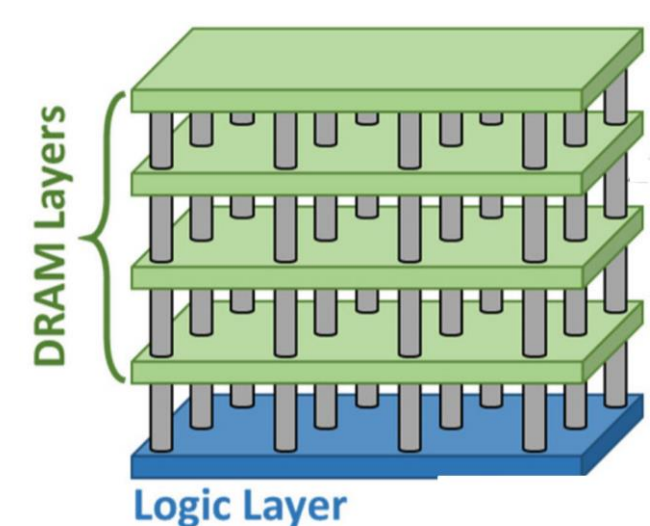
## 1. Motivation

### • Deep neural network (DNN) accelerators designs



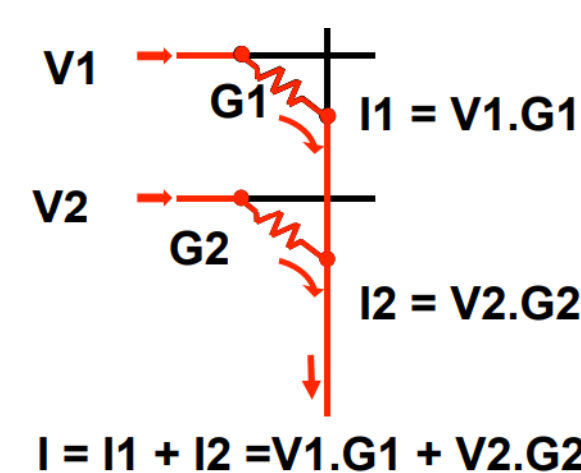
Essential to meet the growing compute demands for high speed and energy efficiency

### • Processing-in-memory (PIM) DNN accelerators



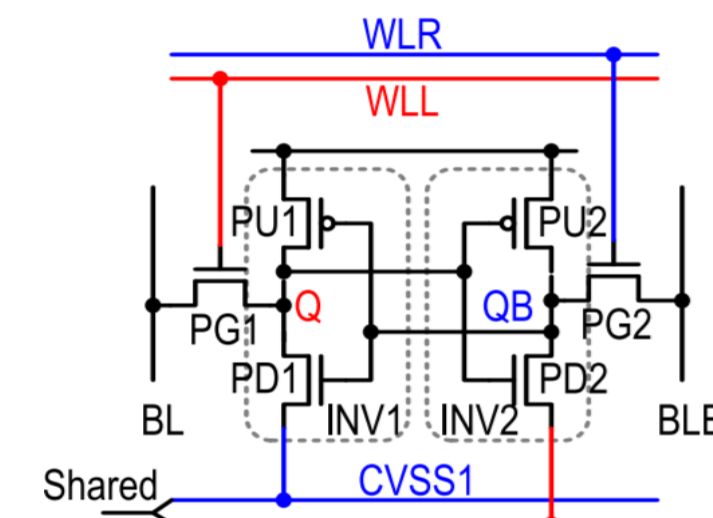
[Ghose, IBM J. Res. Dev. 2019]

DRAM-based



[Shafiee, ISCA 2016]

Memristor-based

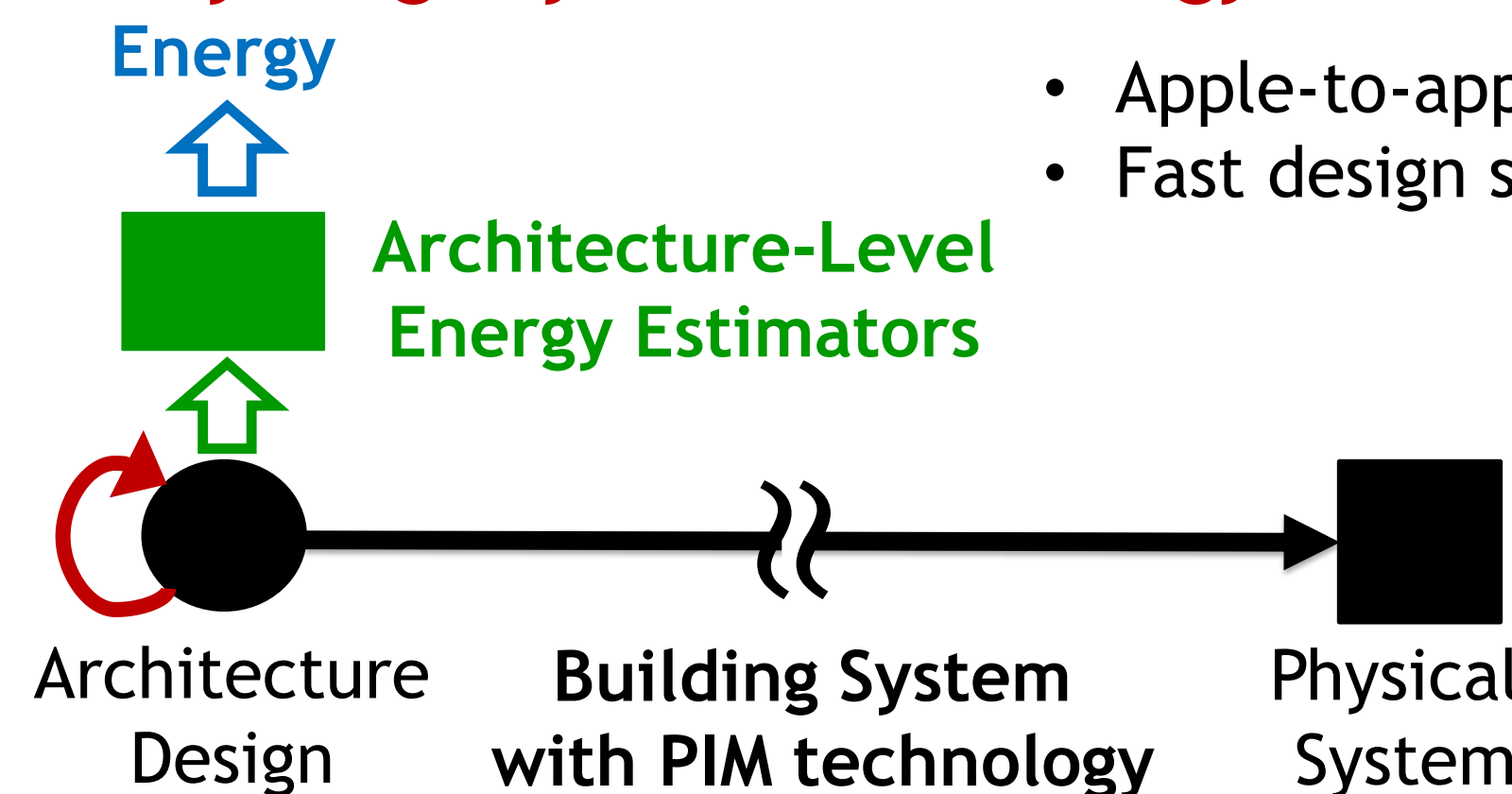


[Si, ISCAS 2019]

SRAM-based

Reduce the amount of data movement by integrating the computation into memory

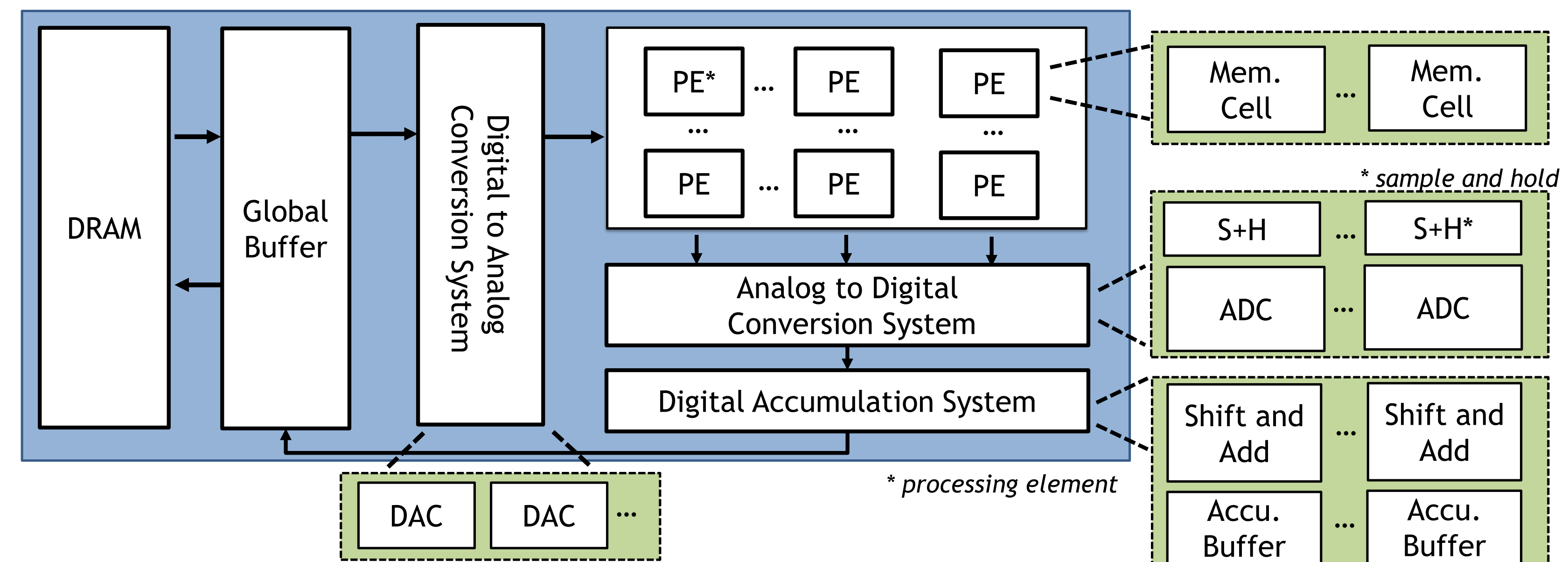
### • Early-Stage system-level energy estimations



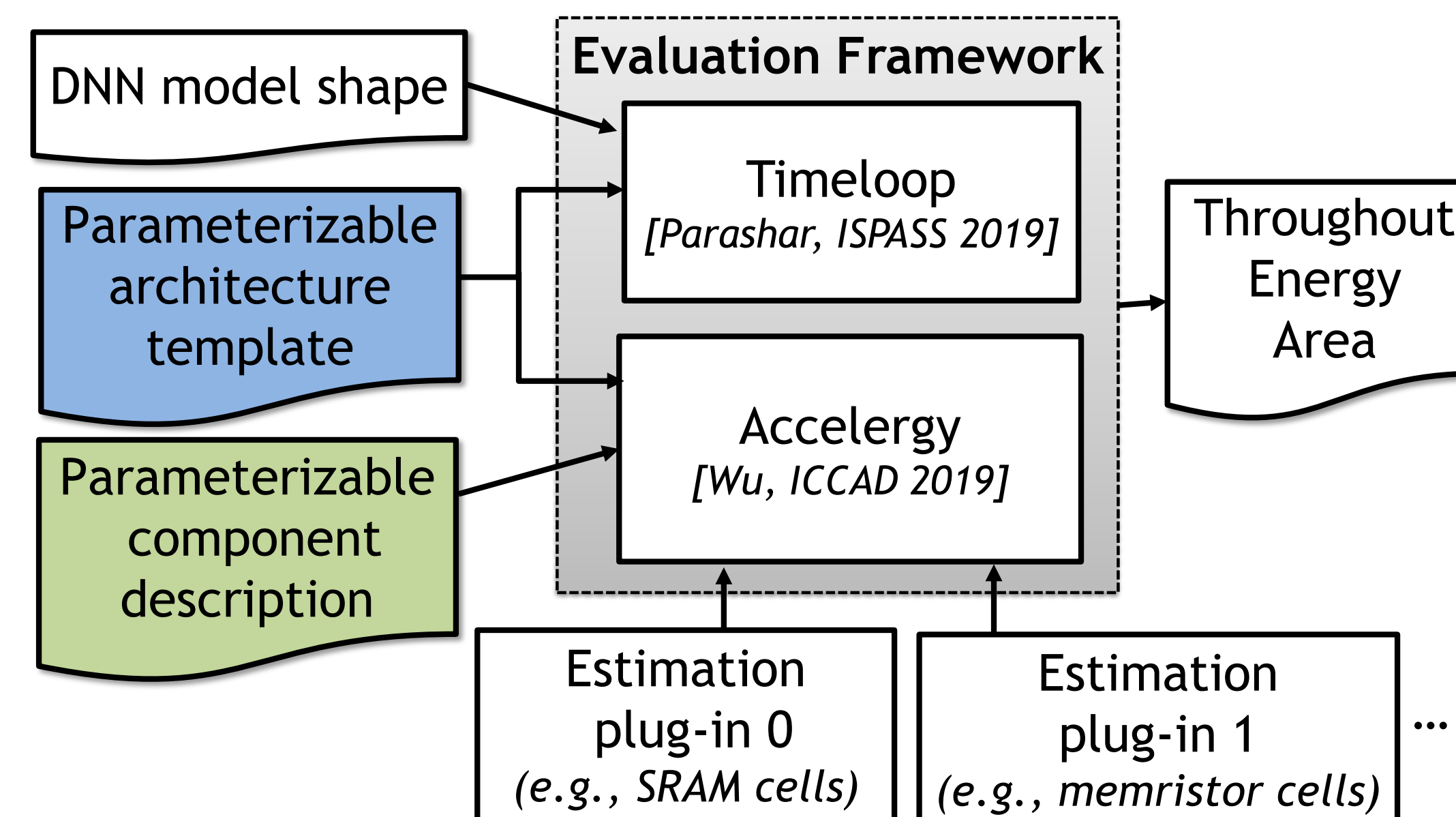
- Apple-to-apple comparisons
- Fast design space exploration

## 2. PIM DNN Accelerator Design Evaluation System

### • General architecture & component template for PIM based DNN accelerators



### • Parameterizable evaluation framework for PIM DNN accelerators



### • Architecture template allows high-level architecture parameter sweeping, e.g.,

- number of PE rows/columns
- size of global buffer, etc.

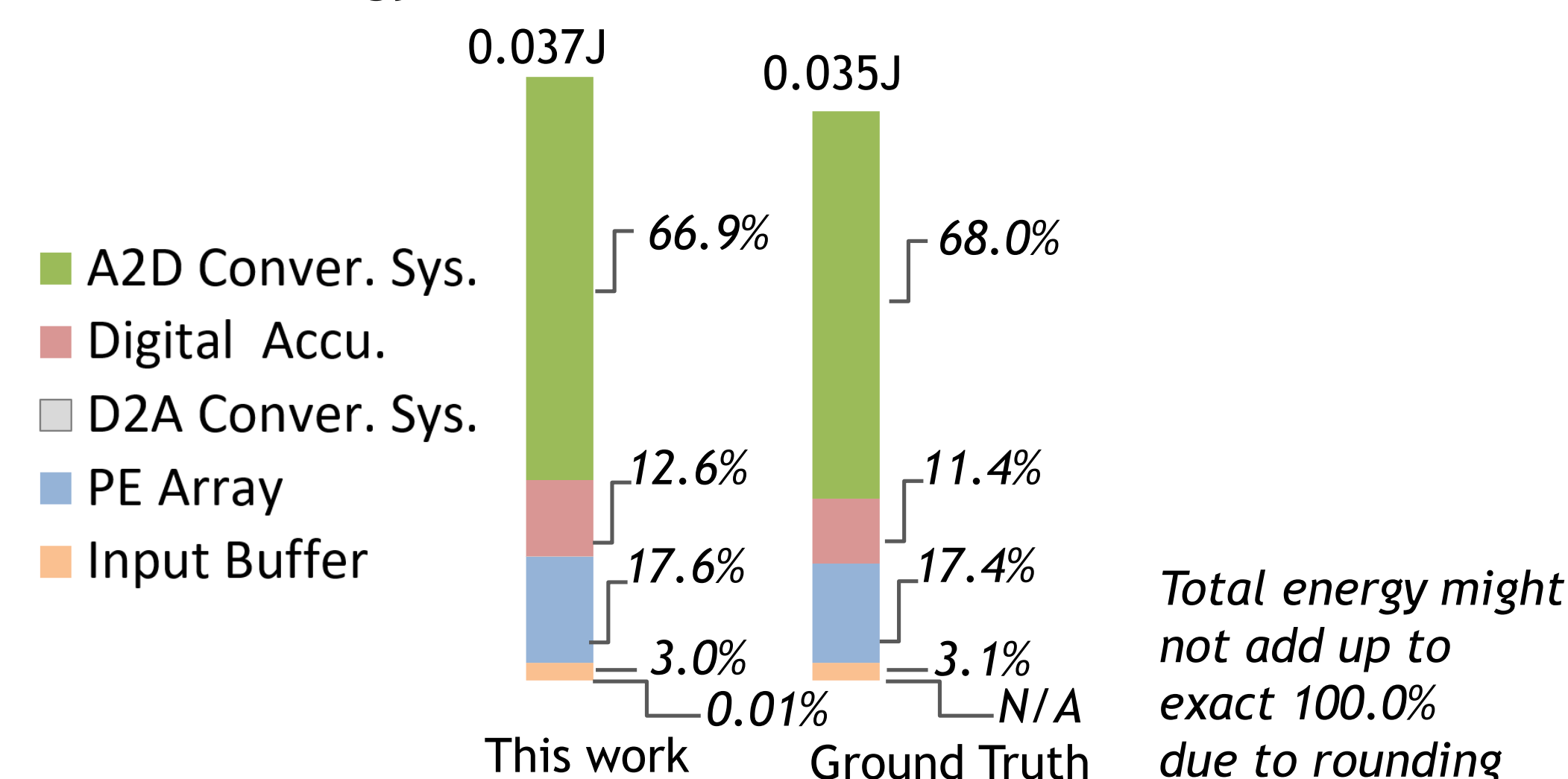
### • Component design template allows lower-level implementation optimization, e.g.,

- optimize DAC-based D2A conversion system
- optimize the design of the flash ADC in the A2D conversion system

## 3. Validation on existing design

- Energy validation performed on the ADC-based design proposed in CASCADE [Chou, MICRO 2019]
- Achieves 95% accurate total energy estimation
- Tracks the breakdown across different components
- Captures the energy breakdown of each convolutional layer

### Total Energy Estimation and Breakdown Validation



### Energy Breakdown Across VGG Convolutional Layers

